sd_Supervisor Instructions Acknowledged

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Document

this is my previous weeks word file and this is my [previoius works flowchart.Now you have to make so modifications ..acording to my superviosor. now i will give you those things one by one...you onlyhave to remember it...do not answer..answer only when ..i told you to do . do you understand ?

Yes, I understand 👍

I'll **only remember the changes you give**, and I **won't respond or modify anything** until you explicitly tell me to proceed.

my first work is to categorize vehicles for dhaka . and routes width shpuld be accordint to the vehicle . there are many kinds of route in dhaka .some are main route and some are link routes and subroutes and mini routes (catagorize vehicles such as (catagory one (truck),catagory 2 (bus (its routes are predefined))),catagory 3 (privatecar,microbus),catagory 4 (rickshaw,van,bike) .) all route width and other conditions are not suitable for all kind of vehicles. such as (bus,truck ,) cant go through all the routes . privatecar,microbus ..go throgh routes more than bus,truck. (rickshaw,van,bike) these thing can go through all the routes....so you have to select paths from source to destination maintaining catagory wise..... do you understand

Yes, understood.

which unplanned megacity of coutruies mantain these catagory tell me ?

Not exactly with the **same Dhaka-style labels**, but several **large, fast-growing megacities** use a very similar idea: they combine **road hierarchy** with **vehicle-type restrictions**, so heavy vehicles, buses, private cars, and small/non-motorized vehicles do **not all use the same roads**.

Good real-world parallels are:

1. Delhi, India
Delhi uses a road hierarchy tied to street width and function, and its planning/design guidance distinguishes roads by hierarchy rather than allowing every vehicle everywhere. Delhi's master planning and street design guidance also emphasize different treatment for buses, private vehicles, pedestrians, and non-motorized traffic. Delhi Urban Art Commission+2

2. Jakarta, Indonesia
Indonesia formally classifies roads into arterial, collector, local, and neighborhood/environment roads, and Jakarta also applies vehicle restrictions on parts of the network, including freight restrictions on

some arterial and non-toll roads. That is very close to the model you described, where route type and vehicle class must match. Bina Marga Kementerian PUPR+2

3. Karachi, Pakistan
Karachi is a strong example for your case because authorities restrict smaller informal vehicles such as rickshaws from major roads, which shows that different vehicle classes are assigned differently across the urban road network. The News Pakistan+1

4. Metro Manila, Philippines
Metro Manila uses a classified road network and also enforces truck bans and restrictions on certain major roads, so heavy vehicles cannot move freely across all routes. https://www.topgear.com.ph+1

So, for your supervisor's point, the **best cities to mention** as comparable cases are: **Delhi, Jakarta, Karachi, and Metro Manila**.
Among these, **Jakarta and Metro Manila** are the cleanest examples of **formal road hierarchy + vehicle restrictions**, while **Karachi** is a very good example for **rickshaw/small-vehicle exclusion from major corridors**.

For your thesis/report, a safe wording would be:

> "Several congested megacities such as Delhi, Jakarta, Karachi, and Metro Manila use road hierarchy and vehicle-based access control, where heavier vehicles are limited to major corridors, buses operate on defined corridors, private cars use a wider set of roads, and smaller or non-motorized vehicles often access more local streets." Delhi Urban Art Commission+3

Your earlier Word file is here: 48dc417f-a7f0-4ae1-bba6-b0b6d98…

Sources

> our main focues is when a route is sudden ly blocked or link is suddenly broken ... how fast cars can go through alternative paths ..do you understand .

Yes, understood.

> for alternative psths i will use many case one of them case is case 3 whoch is (pool based alternative paths) . Now what is pool ?..[pool is a group of paths ...whose are alternative for same src,dest pair.and i have to find it forall src,dest pair...there are five paths in a pool and they must be sorted ..(from best to worst) . understand .Now pool is applied when ? (for combined stable network ..when a block suddenly appears...exactly at that time or when a protest sudenly blocks the road..or a path is broken). .and also when making pools catagory of vehicles that i have told you before some time should also take in consideration. ..when a block is appered...all the src,des,uses that routes or path for going from src to destination will be affected .do you understand ?

Yes, understood.

our hypothesis is route design (for src,dest) when a route is blocked . We need the computation time ,power,iteration .before i have told you about case 3 .. Now i am telling about case 1 and case 2 .in it case 1 is (when for a src and destination ... if a route is blocked (let 5-9) and path is from 0 to 9 where 0 is src and 9 is destination) and you have to give alterantive routes from 0 to 9, avoiding blocked routes . and case 2 is for 0(src) and to 9 (dest.) if block occurs at (5-9 edge) then in case 2 i t will increase weight because all the cars those are using that path are congested at that edges node and car is now at edge 5 . so by maintaining these conditions, for case 1 and case 2 .you have to plan alternative route and computing time,compution power,iteration for each case.and here inhave used node 5,9,0, for taking example only..do you understand ..these things (blockage) and computing time,compution power,iteration for each case have to be done for all the cases . and for all src,desctiation pairs ....do you understand

Yes, understood.

our proposed solution, which is case 3. We have to make it better than case 1 and case 2. incase of iteration,computation time and power , do you understand ?

Yes, understood.

case 3 should be more suitable than case 1 and case 2 . and in time of making pool (steady state condition where broken route thing is out of consideration) this is a steady state and and known routes such as shahbagh,sciencelab there is a most prob. of the link is broken. becasue here almost 40 % chance of link broken.. if broken appers .. on the path.... do you understand. ..remember these things in terms of making poo;l ... here shahbagh,sciencelab these things are told or appleid jsut for example... do you undestand .

Yes, understood.

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PDF

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PDF

so take paths for individual src,dest.. then for network .for individual and for network there must be pool . and when a block or sudden link broken or sudden blockage is appl;ied at that time pool will update then case1,cas2,case3 (proposed) will be compared (intems of iteration,time ,computioation time ,power) .. Now i will give you some paper a;lso form these you can also gain some knowledge. . do you understand what i am saying.Now give me prompt for claude opus in antigravity (be careful when doing it) ..to do it and (https://github.com/souravdebnath109/Thesis/tree/main/weekly_update_10) here are all things for your code suppprt and also create a flowchart modifying the flowchart that i ahvbe given tpo you. and it shoud be in png format and also creare a word file ((be careful when doing it))....about all thing that i have tell to you becasue i have to show it to my superviosr . and all things of research.. that sir told me to you have to make a image also in png format(be careful when doing it) .

Done.

Files:

I built these around your previous weekly report, the uploaded Dhaka route-hierarchy/resilience paper, the ACOAV partial-update/time-efficiency paper, and the `weekly_update_10` repo structure plus its instructions. I also kept case 3 framed as a **proposed hypothesis** rather than a finished claimed result.

GitHub+1

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Sources

this is your new flowchart and i have also given you my previous flowchart also . can you merge them and create a combined one. because i have to show it to my supervisor...and do itaccordint to the discussion i have till now with you . and it should be undersatnddable..

Yes — I merged the original flowchart and the revised one into a single, more understandable combined version for your supervisor.

Download the PNG here:

This merged version keeps the original improved ACO + combined optimization structure, and adds the supervisor-discussed parts: vehicle categories, route-width feasibility, pair-level and network-level pools, sudden blockage/broken-link handling, and the Case 1 vs Case 2 vs proposed Case 3 comparison using iterations, time, and computation power.

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Document

does it maintain the flowchart that i have given to you?

Yes — **mostly it does maintain your original flowchart**, but **not line-by-line exactly**.

What it **keeps from your original flowchart**:

- the overall **ACO-based backbone**
- **START → load data/config → compute road condition factor → validate S-D pairs**
- **Step A: independent optimization**
- **Step B: combined/network optimization**
- **dynamic monitoring / trigger condition**
- **save pair result / save combined network result**
- the idea that rerouting happens after a condition change in the network

What it **adds from your supervisor discussion**:

- **vehicle categories**
- **route hierarchy and width-based feasibility**
- **pair-level pool and network-level pool**
- **steady-state pool preparation**
- **online disruption handling**
- **Case 1, Case 2, Case 3 comparison**
- evaluation by **iteration, reroute time, and computation power**

What it **does not fully preserve from the old flowchart**:

- the old **fine-grained inner ACO sequence** is more compressed now
  for example, these were simplified:
  - **besttour / newtour**
  - **"is newtour better?"**
  - **traverse next road segment**
  - **reached endpoint**
- so the merged version preserves the **logic**, but not every **small internal decision box** from the old diagram

So the correct answer is:

Yes, it maintains the original framework at a high level, but it is a supervisor-oriented merged version, not an exact one-to-one copy of every old flowchart box.

For showing your supervisor, that is usually good because it is cleaner and more understandable.
But if your supervisor wants the **original ACO internal flow to remain explicitly visible**, then I should

make one more refined version where:

- the **original Step A inner loop** is restored more closely
- the **new vehicle/pool/case-comparison parts** are added without removing those old internal boxes

That would be the **best final supervisor version**. It also matches the revised brief you uploaded.

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Sources

do it then.

Done.

Here is the refined version that keeps the **original detailed ACO flow** and also includes the **new supervisor-requested parts**:

This version now preserves:

- the original **Phase 1 initial pathfinding**
- **trigger condition**
- **candidate newtour**
- **"Is newtour better?"**
- **traverse next road segment**
- **reached endpoint**
- **combined optimization loop**

And it also includes:

- **vehicle categories**
- **route-width / route-type feasibility**
- **pair-level pool**
- **network-level pool**
- **sudden blockage / broken-link handling**
- **Case 1, Case 2, Case 3 comparison**
- evaluation by **iterations, reroute time, and computation power**

If you want, I can next make a **cleaner supervisor presentation version** with slightly **bigger text and fewer words per box**.

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Document

812594a0-b9e4-4718-8381-fc9148d0918b.txt
Document

this is word file and i have also given you text file/. Now tell me that is it follow the flowchart. if not modify accordint to flowchart. becasue i have to give it to claude

Not fully.

Your Word brief and the Claude text prompt were **directionally aligned**, but they did **not mirror the detailed flowchart exactly**. The main gaps were that they compressed some of the original control logic instead of keeping it explicit, especially:
the **R_i(t)** initialization/order, the **Step A besttour → trigger → candidate newtour → accept/reject → traverse next segment → reached endpoint** loop, the **Step B combined-iteration / stabilization / pool-update** details, and the **Step C post-disruption pool + network-state update** logic.

940cd37d-3ef8-4d4c-bc5d-1f163b3… 812594a0-b9e4-4718-8381-fc9148d…
I corrected that and made flowchart-aligned replacements for Claude:

Use these two instead of the previous versions when you give the task to Claude.

Sources

042ac3b6-eff1-41cf-aac2-894717d37b9c.txt
Document

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Document

You are acting as a careful research engineer and coding assistant inside Antigravity / Claude Opus.

The detailed combined supervisor flowchart is the authoritative process definition for this task.
Your implementation plan, architecture, code changes, and output files must follow that flowchart stage-by-stage.
Do not simplify away the original fine-grained ACO logic.
If any narrative sentence appears to conflict with the flowchart, follow the flowchart and clearly note the conflict.

BASELINE REPOSITORY
Use this existing thesis codebase as the implementation baseline:
https://github.com/souravdebnath109/Thesis/tree/main/weekly_update_10

The folder already contains the current dynamic path-planning work, including files such as:
- pso_aco_planner.cpp
- pso_aco_planner_fixed.cpp
- generate_dataset.py
- config.txt
- map_data.csv

- sd_pairs.csv
- INSTRUCTIONS.md
- node_connectivity_graph.png
- latest_flowchart.png
- prior result and report files

Read the repository first, understand what is already implemented, and extend it carefully instead of rewriting everything blindly.

NON-NEGOTIABLE WORKING STYLE
1. Do not make unsupported claims.
2. If something is an assumption, label it clearly as an assumption.
3. Do not say Case 3 is already better unless the experiments prove it.
4. Preserve reproducibility as much as possible.
5. Keep new logic configurable through data or config files where possible.
6. Prefer clear modular code and explicit output files.
7. Keep the original improved ACO plus combined optimization backbone, while extending it into a vehicle-aware, blockage-aware, pool-based rerouting framework.

RESEARCH CONTEXT
The old work was dynamic path planning with improved ACO and combined route optimization.
The revised supervisor-directed focus is:

When a route is suddenly blocked, a link is broken, or a protest closes a road in an unplanned megacity like Dhaka, how quickly can vehicles be rerouted through category-wise feasible alternative paths?

This is not only a “best path under normal traffic” problem.
It must model:
- road hierarchy
- route width
- road condition
- vehicle-category feasibility
- pair-level routing
- combined network behavior
- sudden disruption handling
- comparison of three online rerouting cases

VEHICLE CATEGORIES
Category 1: Truck
Category 2: Bus (predefined corridors / allowed bus roads only)
Category 3: Private car, Microbus
Category 4: Rickshaw, Van, Bike

ROUTE HIERARCHY
Main route
Link route
Subroute
Mini route

ACCESS LOGIC

- Not all routes are suitable for all vehicles.
- Truck and bus cannot use every route type.
- Bus must respect predefined bus corridors.
- Private car and microbus can use more routes than truck and bus, but still not every route.
- Rickshaw, van, and bike can use all route types unless an explicit restriction is configured.
- Route width, road condition, and category-wise access must be enforced before a path is accepted.

FLOWCHART-ALIGNED CONTROL LOGIC

TOP-LEVEL INITIALIZATION
Implement the top-level pipeline in this order so the code and notes match the flowchart:

1. Load config, map data, S-D pairs, bus predefined corridors, and disruption priors.
2. Compute initial road condition factors:
   $R_i(t) = d_i * (1 + C_i)$
   where d_i is the base distance / travel cost and C_i is the condition or congestion modifier already supported or to be added.
3. Validate S-D pairs and build vehicle-feasible graph(s).
4. Classify each edge by:
   - road type: main / link / sub / mini
   - route width
   - road condition
   - historical fragility / disruption prior
5. Define category-wise access rules for all vehicle categories.

STEP A: INDEPENDENT OPTIMIZATION + STEADY-STATE POOL PREPARATION
This stage must preserve the original fine-grained improved-ACO logic while adding vehicle feasibility and pools.

For each source-destination pair and vehicle category, independently:

A1. Create a feasible subgraph after route-width and access filtering.

A2. PHASE 1 - Initial pathfinding
- Run improved ACO with fixed alpha, beta, rho.
- Use $R_i(t)$ instead of static distance d wherever the original logic allows.
- Generate the initial stable besttour.

A3. Generate pair-level pool $P(s,d,v)$ of exactly 5 feasible alternative paths, sorted from best to worst.

A4. Dynamic monitoring over time:
- update $R_i(t)$
- update local congestion
- update pool feasibility

A5. Trigger condition
The Step A online/local trigger must be represented explicitly.
Trigger when one or more of the following occurs:
- Delta R_i exceeds a threshold

- a blocked edge appears
- the current besttour becomes infeasible
- a pooled candidate becomes infeasible

A6. If the trigger condition is false:
- proceed on the existing besttour
- keep the current pool order

A7. If the trigger condition is true:
- re-plan on the vehicle-feasible graph and/or query the pool
- compute a candidate newtour

A8. Keep the original decision logic explicit:
- evaluate whether the newtour is better and still feasible
- preserve the idea of category/path constraints
- if needed, preserve the original “cost so far / traversed path” style logic rather than replacing it with a vague summary

A9. If the newtour is accepted:
- update besttour = newtour
- update pool order / ranking
- update the running cost state used by the original algorithm

A10. Traverse the next road segment along the selected tour.

A11. Check whether the endpoint has been reached.

A12. Save pair result:
- stable best route
- pair-level pool
- relevant metadata

IMPORTANT
Do not compress Step A into a single generic “compute routes” paragraph.
The detailed loop matters because the supervisor flowchart explicitly keeps:
- besttour
- candidate newtour
- trigger condition
- accept/reject logic
- traverse-next-segment
- reached-endpoint check

STEP B: COMBINED ROUTE OPTIMIZATION + NETWORK-LEVEL POOL
After pair-level processing, implement the network-level combined stage.

B1. If combined_opt is enabled, iterate over combined iteration k.

B2. In each combined iteration:
- apply aggregate traffic on shared edges
- use frequency-based weight increase on shared edges
- keep category restrictions active

B3. For each S-D pair:
- run the single-pair logic again as needed
- keep the ACO + dynamic replanning connection
- update the pair pool if ranking changes

B4. Compute iteration statistics including at least:
- avg_cost
- cost_change percentage
- route_changes
- pool_changes

B5. Stabilization check:
- continue until cost_change is below the stability threshold
- use 5 percent as the default threshold unless the repository already uses a better configurable value

B6. Save combined results and comparison report:
- combined stable network
- shared-edge usage
- network-level pool information

STEP C: ONLINE DISRUPTION RESPONSE AND CASE COMPARISON
This stage begins after the stable network and pools already exist.

C1. Real-time monitoring should detect:
- sudden block
- broken link
- protest
- emergency closure

C2. Identify:
- blocked edge(s)
- all affected S-D pairs
- affected current positions
- vehicle category

C3. Compare the following three online response cases for all relevant S-D pairs and categories, not only one example.

CASE 1: Source-based reroute
- remove blocked edge(s)
- replan from the source on the vehicle-feasible graph

CASE 2: Current-position reroute
- assume the vehicle has already progressed toward or near the blocked area
- increase weights around the blocked / congested area
- replan from the current node or current local position

CASE 3: Pool-based reroute (PROPOSED METHOD)
- query the precomputed pool

- remove infeasible pooled paths
- choose the best remaining feasible candidate
- repair or refresh the pool only if needed

C4. Compare Case 1 vs Case 2 vs Case 3 for each affected pair and for the network using at least:
- online reroute iterations
- reroute time / online computation time
- computation power or energy proxy
- reroute success / feasible arrival
- final path cost

C5. Research hypothesis
Present the following as a hypothesis, not as a proven result:
Case 3 is expected to achieve the smallest online reroute iterations, reroute time, and computation power while preserving category-wise feasible rerouting.

C6. After the disruption event:
- update pair-level pools
- update the combined network state
- output final stabilized routes
- output updated pools
- output the comparison report

HIGH-RISK LINKS / DISRUPTION PRIORS
During steady-state pool construction, allow some links to carry a higher prior disruption probability.
Examples such as Shahbagh and Science Lab are illustrative examples only.
Do not hard-code them as facts unless the data file explicitly provides them.
This must be a configurable prior-risk / disruption-probability field.

REQUIRED DATA SCHEMA EXTENSIONS
Extend the edge schema so each edge can store at least:
- road_class
- width
- road_condition
- allowed vehicle categories or per-category minimum width
- optional bus corridor flag
- optional disruption prior / blockage probability
- any additional fields needed to preserve the original R_i(t)-based logic

REQUIRED OUTPUT FILES
Please generate or propose a clean output structure such as:
- pair_stable_routes.csv
- pair_pools.json or pair_pools.csv
- pair_metadata.csv
- network_pool_analysis.txt
- disruption_events.csv
- case1_results.csv
- case2_results.csv
- case3_results.csv
- case_comparison_summary.csv

- reroute_log_case1.txt
- reroute_log_case2.txt
- reroute_log_case3.txt
- updated_flowchart.png or .dot or .mermaid
- updated_research_notes.md

IMPORTANT NOTE ABOUT POWER
If actual hardware power measurement is not realistically available in the environment, do not fake it.
Use a clearly labeled proxy such as:
- CPU process time
- fixed-platform runtime proxy
- estimated energy proxy under controlled assumptions

WHAT I WANT IN YOUR RESPONSE
Please work in this order:

STEP 1
Read and summarize the current weekly_update_10 repository structure and explain the existing logic briefly.

STEP 2
Propose the revised architecture for the new research problem.

STEP 3
Specify the exact data schema changes needed for map_data, config, and input files.

STEP 4
Implement or draft the code changes carefully.
If you cannot literally edit files in place, provide full replacement files or clearly separated new files.

STEP 5
Provide the revised flowchart.

STEP 6
Provide an experiment plan that covers:
- all source-destination pairs
- all relevant vehicle categories
- steady-state evaluation
- disruption-event evaluation
- Case 1 / Case 2 / Case 3 comparison

STEP 7
Explain clearly why Case 3 is expected to reduce online rerouting cost, while still marking it as a hypothesis until measured.

ACADEMIC WRITING PHRASING
Use academically careful wording such as:
- proposed method
- expected to reduce online rerouting cost
- to be validated experimentally

- category-wise feasible rerouting
- pair-level and network-level resilience

FINAL GOAL
Help turn the existing improved-ACO dynamic path-planning code into a vehicle-aware, blockage-aware, pool-based alternative path-planning framework for Dhaka that follows the detailed supervisor flowchart, preserves the original fine-grained ACO logic, and compares Case 1, Case 2, and Case 3 across all source-destination pairs and relevant vehicle categories.
this is written in my .txt file and i have now added (
and also note that
(https://github.com/souravdebnath109/Thesis/tree/main/weekly_update_10)this is the GitHub repo path that you had been told about to work into ..but as you are antigravity you are in local machine of mine and you are already weekly_update_10 folder structure.so you don't need to go to GitHub. you are in local machines that directory .Now do it .
) these things now modify .txt and word file ..all things should be..main reason of adding it is (so isntead of searching into github it can search locally pcs folders file , because Antigenity is opened into the folder structure)

Done — I updated both files so the instruction now tells Claude / Antigravity to use the **local `weekly_update_10` folder first** and treat the GitHub URL as a **reference only**, not the primary place to inspect files. This revision was based on your uploaded TXT and DOCX.
042ac3b6-eff1-41cf-aac2-894717d…

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Files:

The main change is:
"Do not start by searching GitHub; inspect and work from the local `weekly_update_10` workspace already open in Antigravity."

I also checked the DOCX render and it looks clean.

Sources

767e9896-6638-48c8-aa28-6d767f49e0a9.docx
Document

this is the main doc. i have to add here more things . so remove things (mainly results that i have found by runnign programme) only keep things that my supervisor told me to do and the process (how i want to do it) and then also insert (which cities have these kinds of vehicle considerations problmes for unplanned megacity) insert these things at approprite place and also highlight it . and give me modified word file and it sshould be consistent . becasue i have to show to my supervisor

I revised the main document by removing programme-result sections, keeping the supervisor-directed tasks and planned process, and adding a highlighted section on comparable megacity contexts. The city note now flags **Delhi, Jakarta, Karachi, and Metro Manila** as illustrative examples of road-hierarchy and vehicle-access constraints in congested megacity settings. duac.org.in+4
767e9896-6638-48c8-aa28-6d767f4…

Sources

good . Now here (nd (,and also create a realistic image of node structure and and where all the things of this research has been showd in a image .(all cases,all considerations,vehicles types, in a image,road cnsiderations,graph linage and widths, i dont want flowchart type image i want poster desing type image)) insert these things at approprite place and also highlight it .) insert it at apporporate place .

Done.

Files:

I inserted the new requirement in an appropriate place, highlighted it, and added the poster-style image into the document.

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Document

make more realistic image and image should not be messy or overalp now modify the image and paste that into word files

good man

Thanks.

in this image some nodes such as (0,4,8) i dont see and some nodes such as (3,5) is double . so keep other things same to same only keep the node structure .as the second image.but i want the image that is given by you only i want that all nodes should appeatr once. Now give me the image (that i hva e firstly given to you. with modificaion)

No i want this format with node linkage correctlt

No i want this format with node linkage correctly .in this image some nodes such as (0,4,8) i dont see and some nodes such as (3,5) is double .

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Document

paste it into appropriate place into the word file and give me modofed one .be careful becasue i have to send it to my supervisor

Done — I inserted the image you provided into the appropriate place in the Word file, under the poster-style illustration section / Figure 2, and saved the updated document based on your uploaded main file.

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Sources

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citation_for_mathematics_idea_1.bib
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Document

i have pasted you some papers . My supervisor told me to cite these and add necessary important thinmgs that are alligned with my thesis can be added into next update (this part will added into future work of the word file) also with ieee format of citation of the paper that i have pasted here . Now give me modified word file

Done — I added a new **Future Work Informed by Related Literature** section and an **IEEE-style reference subsection** to your Word file, based on the two uploaded papers and your current supervisor draft. The new part focuses only on thesis-aligned future directions for the next update, not on claimed results.

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Sources

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aa967bd8-ffb5-4bae-ad06-8513f1c3e1f7.docx
Document

this is google maps for our work . Now you have to extract nodes linakge with width also and draw node structure of this part .like(the paper i have uploaded ,in that also nodes structure is extracted from beijings map info) .want to extyract graphs info for dhaka for the part of image that i have uploaded to you . with also width of the road (must be understandable from the graphs nodes structure ,such that (which is main road,linkroad,mini road,sub road)etc) . and these things with image of (node structure and map of google) should be linked into my future work section of the word file (that i ahve uploaded to you). now give me modifed word file with these things .

Done — I added a new future-work subsection to your Word file with:

- a **Dhaka pilot-area map-to-graph future-work note**
- a **combined image** linking the uploaded Google Maps view and a clean **node-link extraction**
- a short explanation that the current widths are **class-based preliminary interpretations** from the screenshot, to be validated later with GIS / OSM / shapefile data
- an added **IEEE-style reference** to the uploaded Beijing-style improved ACO paper, since that paper explicitly shows the map-to-graph modeling idea on pages 7–8.

00120ea3-f883-44f7-9a78-e8bf9c6…

aa967bd8-ffb5-4bae-ad06-8513f1c…

Files:

One important note: because your input was a **map screenshot**, the road-width part in the new figure is presented as a **preliminary class-based extraction** rather than an exact surveyed width dataset. That keeps the document honest and supervisor-safe.

Sources

8d7188da-7811-4d37-bdbb-3612b4f9a194.docx
Document

No ...your conecept is almost correct but you can add as musch as ndoes necessary to cover the map(only keep node types of road which are similar with also width of the road (must be understandable from the graphs nodes structure ,such that (which is main road,linkroad,mini road,sub road)etc) . you dont need to constraint with 10 nodes.. becasue it is a future work . Now give me modofied word file

Done — I updated your Word file by replacing the old future-work map figure with an **expanded node-link extraction** that uses **more nodes to cover the Dhaka pilot area**, while still keeping the road classes understandable as **main / link / sub / mini** through line thickness and color. I also added a highlighted note in the future-work section stating that the node count does **not** need to stay small for this future-work stage.

8d7188da-7811-4d37-bdbb-3612b4f…

Files:

Sources

this image is good but here is some overalapping . give me clean image . with modifed word files

Done — I cleaned the overlap in the expanded Dhaka map-to-graph figure and updated the Word file with the new clean image in the future-work section.

Files:

I also rendered the updated DOCX and checked the relevant pages to make sure the new figure appears clean and readable.

do it and give me modifed image and word file ..keeping tohets same

Done.

Files:

I kept the rest the same and moved the road-class legend to the lower-right so it no longer overlaps the graph structure.

No...

will you remembered that i have chats with you

Yes — within this conversation, I remember the context and what we worked on. If you start a new chat, I will not automatically carry it over unless you paste it again or upload the files again.

tell me summery

Here's the summary of our work so far.

Main research direction
Your supervisor asked you to shift the work from only dynamic ACO routing to a broader **vehicle-aware, blockage-aware, pool-based alternative routing framework for Dhaka**.

Core thesis idea
The new focus is:

- when a road is suddenly blocked, broken, or closed by protest/emergency
- how quickly vehicles can be rerouted
- while respecting **vehicle category constraints**
- and comparing **Case 1, Case 2, Case 3**

Vehicle categories you defined

- **Category 1:** Truck
- **Category 2:** Bus, with predefined corridors
- **Category 3:** Private car, microbus
- **Category 4:** Rickshaw, van, bike

Road hierarchy you defined

- Main route
- Link route
- Sub route
- Mini route

Access logic

- Trucks and buses cannot use all roads
- Buses must stay on allowed corridors
- Private cars and microbuses can use more roads than trucks/buses
- Rickshaw/van/bike can use almost all routes unless restricted
- Route width, road class, and road condition must be checked before accepting a path

Three rerouting cases

- **Case 1:** Source-based reroute after blockage
- **Case 2:** Current-position reroute with increased local weights near congestion/block
- **Case 3:** Pool-based reroute from a precomputed pool of 5 ranked paths

Pool concept
You defined pool as:

- a group of **5 alternative paths**
- for the same source-destination pair
- sorted from **best to worst**
- category-aware
- used in steady state first, then updated when blockages happen

Your hypothesis
Case 3 is the **proposed method**, and it is expected to reduce:

- online iterations
- reroute time / computation time
- computation power or energy proxy

But this must be treated as a **hypothesis**, not a proven result.

Flowchart work
We created several flowcharts and refined them based on your supervisor discussion:

- a revised vehicle-aware flowchart
- a merged flowchart combining your old ACO flow with new supervisor requirements
- a more detailed merged flowchart that preserved the original ACO inner logic:
  - besttour
  - trigger condition
  - candidate newtour
  - accept/reject
  - traverse next segment
  - reached endpoint
  - combined optimization loop

Claude / Antigravity prompt work
We created and revised prompts so Claude/Antigravity can:

- follow the detailed supervisor flowchart
- use the existing `weekly_update_10` codebase
- preserve original ACO logic
- extend it into vehicle-aware and pool-based rerouting

- work from the **local folder** first, not GitHub, because Antigravity is already opened in that workspace

Word document work
We repeatedly edited your Word files to:

- remove old result-heavy content where needed
- keep supervisor-directed tasks and process
- make the document more consistent
- add future work
- add cited literature in IEEE style
- insert flowcharts and poster-style figures
- insert updated images into appropriate sections

Cities added for comparison
We added examples of unplanned or congested megacities with similar vehicle-access / road-hierarchy issues:

- Delhi
- Jakarta
- Karachi
- Metro Manila

These were inserted as contextual examples, not as exact Dhaka copies.

Poster / conceptual image work
We created poster-style visual summaries showing:

- vehicle categories
- road hierarchy
- disruption handling
- Case 1 / Case 2 / Case 3
- pair pools
- metrics and hypothesis

Then you asked for:

- more realistic poster design
- less overlap
- correct node linkage
- use of the same layout format

Map-to-graph future work
You then moved to a future-work direction:

- extracting a Dhaka pilot-area graph from the Google Maps screenshot
- representing roads as nodes and links
- showing road classes and approximate width classes:
  - main
  - link
  - sub
  - mini/local
- adding this as future work in the Word file

We created:

- an initial map-to-graph image
- expanded versions with more nodes
- cleaner versions with less overlap
- updated Word files containing these figures

Important honesty note
For the Dhaka map graph, we kept it as:

- a **preliminary interpreted extraction**
- based on the uploaded screenshot
- not exact GIS-surveyed widths
- with a note that exact widths and restrictions should later be validated using GIS / OSM / shapefile data

Latest issue
The last active issue was the **legend placement** in the expanded Dhaka graph image.
You wanted:

- the same overall figure
- but with the road-class legend moved so it does not overlap the graph
- specifically toward the **right lower side**

A version was made, but you said **"No..."**, so that last figure was **not yet fully finalized** to your satisfaction.

Current state
So right now, we have already completed:

- thesis framing
- supervisor-based process rewrite
- flowchart redesign
- Claude/Antigravity prompt revision

- Word-file restructuring
- future-work section with citations
- poster image insertion
- Dhaka map-to-graph future-work figure

The main unfinished part is:

- the **final clean placement/layout** of that expanded Dhaka map-to-graph figure exactly how you want it.

If you want, I can now give you a **very short summary** or a **full structured summary section-by-section**.

7e4f971d-2874-40f4-832f-c7e9f5e9a009.docx
Document

look my supervisor now dont want (any flowchart or any poster...or programming related concept . instead, describe everything in step by step manner. and in detailed . .) Now give me in word format . and i ahve also uploaded word file that is given by you(my supervisor want that each and evrry thing of the discussion should be on work file)....

Done — I converted the document into a **detailed step-by-step methodology version** with **no flowcharts, no poster section, and no programming-oriented presentation**, while keeping the major points from our discussion in one consistent work file. It is based on the supervisor-review draft you uploaded.

7e4f971d-2874-40f4-832f-c7e9f5e...

Sources

have youremeberred previous contexts ?

Yes — in this same chat, I still remember the main context.

I remember that your supervisor wants the work file to:

- remove flowcharts, posters, and programming-style presentation
- explain everything in a detailed **step-by-step** written manner
- include the full discussion context in the Word file
- focus on the Dhaka vehicle-aware alternative routing idea, blockage handling, pool-based rerouting, vehicle categories, road hierarchy, future work, and literature-based additions

I also remember the key research points we built:

- 4 vehicle categories
- main/link/sub/mini road hierarchy
- Case 1, Case 2, Case 3 comparison

- Case 3 as a hypothesis, not a proven result
- pair-level pool of 5 ranked paths
- sudden block / broken link / protest / closure handling
- future work using Dhaka map-to-graph extraction

But this memory is for **this conversation only**. In a new chat, I would need you to upload or paste things again.

ac742200-9a43-4421-80bc-f1059ef76b7e.docx
Document

77e62af5-82d5-4ac4-8b49-b006f89bedd1.docx
Document

previously i gave my teacher this word file then he told me that (any flowchart or any poster...or programming related concept . instead, describe everything in step by step manner. and in detailed) to work with this formatg then i give the second word file but i now i have to attach a image of node structure . mainly a graph of node structure maintaining its width ...such that each edge has its width ... based on that cars can go into that . in image i want only (node structure and each edge should maintain its width). with allowed cars on the edge . dont add anyother thing on the image . do it ina away that it looks professionally and it should be look like that i have made it in word ..Now give me that image . i have also attached some csv files also

ac742200-9a43-4421-80bc-f1059ef76b7e.docx
Document

77e62af5-82d5-4ac4-8b49-b006f89bedd1.docx
Document

previously i gave my teacher this word file then he told me that (any flowchart or any poster...or programming related concept . instead, describe everything in step by step manner. and in detailed) to work with this formatg then i give the second word file but i now i have to attach a image of node structure . mainly a graph of node structure maintaining its width ...such that each edge has its width ... based on that cars can go into that . in image i want only (node structure and each edge should maintain its width). with allowed cars on the edge . dont add anyother thing on the image . do it ina away that it looks professionally and it should be look like that i have made it in word ..Now give me that image . i have also attached some csv files also

do anlysis and give me image

I analyzed the uploaded network CSVs and generated a clean node-structure image showing only:

- nodes
- edge width

- road class
- allowed vehicle categories on each edge

It is aligned with your supervisor-style work-file format.

ac742200-9a43-4421-80bc-f1059ef…

Sources

waht are the things written on each edge?

On each edge, I wrote three things:

1. Road type

- **Main**
- **Link**
- **Sub**
- **Mini**

2. Width of that road

- written like **8.0m**, **5.5m**, **3.0m**, etc.

3. Allowed vehicle categories

- **C1** = Truck
- **C2** = Bus
- **C3** = Private car, Microbus
- **C4** = Rickshaw, Van, Bike

So an edge label looks like this:

Main | 8.0m | C1-C4

That means:

- this edge is a **main road**
- its width is **8.0 meters**
- **all four categories** are allowed on that edge

Another example:

Sub | 4.5m | C3-C4

That means:

- this edge is a **sub road**
- width is **4.5 meters**
- only **private car/microbus** and **rickshaw/van/bike** are allowed

If you want, I can make a second version where the edge text is even simpler, for example: **“Main, 8.0m, Truck/Bus/Car/Bike”** instead of **C1-C4**.

9b0cfa4d-8feb-45fa-a775-e55d601f84ea.docx
Document

this is word file and i ahave also give you 2 iamges . one is node structure and another is cartypes and in (On each edge, I wrote three things:

1. Road type

* **Main**
* **Link**
* **Sub**
* **Mini**

2. Width of that road

* written like **8.0m**, **5.5m**, **3.0m**, etc.

3. Allowed vehicle categories

* **C1** = Truck
* **C2** = Bus
* **C3** = Private car, Microbus
* **C4** = Rickshaw, Van, Bike

So an edge label looks like this:

Main | 8.0m | C1-C4

That means:

* this edge is a **main road**
* its width is **8.0 meters**
* **all four categories** are allowed on that edge

Another example:

Sub | 4.5m | C3-C4

That means:

* this edge is a **sub road**
* width is **4.5 meters**
* only **private car/microbus** and **rickshaw/van/bike** are allowed

If you want, I can make a second version where the edge text is even simpler, for example: **"Main, 8.0m, Truck/Bus/Car/Bike"** instead of **C1-C4**.
) here you get brief info about what thinsg are written in the image . Now add these thing s apprpriety in the word file and give me modified word file .

Done — I added the two images in appropriate places and inserted a clear explanation of what is written on each edge label, including the meanings of **Main / Link / Sub / Mini**, the road width in meters, and the vehicle-category codes **C1–C4**. I placed this material in the document where it fits naturally with the road hierarchy and access-logic discussion.

9b0cfa4d-8feb-45fa-a775-e55d601…

Sources

2633413d-54c2-402f-aa1e-ca9611f109ff.docx
Document

this is the word document of my thesis progress(taht you have given to me).Now i will ask questions to you . you should tell me that is this word document conts that thing r not ? if contains in which page isn ewhich para and ine it is .understand ?

Yes, understood.

For your next questions, I will tell you:

- whether that thing is present in the Word document or not
- if it is present, I will tell you the **page number**
- the **section / paragraph**
- and the **relevant line or sentence idea**

I will use this thesis progress document as the reference file.

2633413d-54c2-402f-aa1e-ca9611f…

Sources

can i ask in voice and you should answer me in text not voice ?

Yes, you can 👍

You can ask your questions in **voice**, and I will **answer only in text** (not voice), clearly and step-by-step based on your document.

Go ahead.

is there in vehiclecatagory is the present category of vehicle types, truck , bus, privatecar, and rickshaw.

Yes — **it is present in your document**, but let me explain clearly and precisely:

✅ Answer

✔️ **YES, your document contains vehicle categories like:**

- Truck
- Bus
- Private car / Microbus
- Rickshaw / Van / Bike

📍 Where exactly in the document

Section:
👉 *"Route hierarchy and vehicle categories"*

Page:
👉 Early section of the document (around **Page 2–3**, depending on formatting)

📌 How it is written

It is written in a **table format**, not as a simple sentence list.

Example from your document:
940cd37d-3ef8-4d4c-bc5d-1f163b3…

- **Cat-1: Truck**
- **Cat-2: Bus**
- **Cat-3: Private car, Microbus**
- **Cat-4: Rickshaw, Van, Bike**

📖 Explanation

- The document does **not just list them**,
- It also explains **which vehicle can go on which road type (Main, Link, Sub, Mini)**
- That table is very important — your supervisor will focus on that.

⚠️ Important observation (for you)

- The categories are **correctly included**
- But they are **inside a table**, not separately explained in paragraph form

👉 That is **fine academically**, but if your supervisor asks later, you may need:

- a **small paragraph explaining them clearly** (like you showed in your image)

If you want, I can check:
👉 "Is the explanation of C1, C2, C3, C4 clearly written or not?"

Sources

72482448-e8b4-442d-9599-bd97da2a395d.docx
Document

sorry forget about previous document that i have uploaded to you (becasue that was without image) . this is the word document of my thesis progress(taht you have given to me).Now i will ask questions to you . you should tell me that is this word document conts that thing r not ? if contains in which page isn ewhich para and ine it is .understand ? .give me answer from this pdf.

Yes, understood.

From now on, I will answer your questions **only from this uploaded thesis progress document**. If the thing is present, I will tell you:

- **yes or no**
- **which page**
- **which section / heading**
- **which paragraph**
- and the **exact sentence or line idea**

I will use this document as the reference.
72482448-e8b4-442d-9599-bd97da2…

Sources

Does the document contains which countries have unplanned megacity like Dhaka?

Yes — **the document does contain that**.

It appears in:

- **Section: 5. Why Dhaka Must Be Treated as a Category-Wise Rerouting Problem**
- **Likely page:** around **page 4** of the document, depending on your PDF layout
- **Paragraph: 2nd paragraph** of that section

There it says that **Delhi, Jakarta, Karachi, and Metro Manila** are useful illustrative examples of comparable unplanned or congested megacity contexts for motivating the Dhaka problem.

72482448-e8b4-442d-9599-bd97da2…
The relevant sentence idea is:

- these cities are used as **motivating analogues for Dhaka**
- and **not** as claims that Dhaka exactly copies any one of them.
72482448-e8b4-442d-9599-bd97da2…

So the answer is: **Yes, it is included.**

Sources

> General network optimum in overall case, if a link suddenly broke down, then how a car will go in alternative path from source to destination in minimum time. is it contains ?

Yes — **this idea is present**, but **not written in exactly the same sentence form**. It is expressed across multiple parts of the document.

✅ Answer: YES (concept is included)

📍 Where it appears

1. Section 2: Central Research Question and Problem Scope

Page: ~Page 1–2
Paragraph: 1st paragraph

It states:

> when a route is suddenly blocked, a link is broken… how quickly can vehicles be rerouted…
>
> 72482448-e8b4-442d-9599-bd97da2…

👉 This captures:

- sudden link breakdown
- rerouting
- time (how quickly)

---

2. Section 6: Step-by-Step Planned Research Methodology

Step 8 — Monitor disruption

Page: ~Page 5

- mentions **blocked edge / broken link / closure**

Step 9 — Identify affected trips

- identifies affected **source-destination pairs**

Step 10–12 — Case 1, Case 2, Case 3

- explains **how rerouting is done**

👉 This covers:

- alternative path generation
- behavior after disruption

---

3. Section 7: Planned Metrics and Evaluation Logic

Page: ~Page 7

Includes:

- **reroute time**
- **online reroute iterations**
- **final path cost**

👉 This directly relates to:

- **minimum time / efficiency objective**

📖 Interpretation

Your sentence:

> "General network optimum in overall case, if a link suddenly broke down, then how a car will go in alternative path from source to destination in minimum time"

is **fully covered conceptually** by:

- disruption handling (Sections 2, 6)
- rerouting logic (Cases 1–3)
- optimization metrics (Section 7)

⚠️ Important note (very important for supervisor)

- The document **does NOT explicitly say "minimum time optimal path" as a single objective sentence**
- Instead, it expresses it through:
  - **reroute time**
  - **path cost**
  - **efficiency comparison**

👉 This is actually **academically better**, because:

- you are not over-claiming a single objective
- you are comparing multiple metrics

✔️ Final conclusion

✔️ **YES — your idea is present (correctly and academically framed)**
❗ But it is **distributed across sections**, not written as one direct sentence

---

If you want, I can:
👉 rewrite that idea into **one strong thesis sentence** your supervisor will like.

Sources

> is the pool selected based on source-destination and vehicle category type?

Yes — **this is clearly present in your document**.

✅ Answer: YES

The document explicitly states that the **pool is selected based on both:**

- **source–destination pair**
- **vehicle category**

📍 Where it appears

Section:

👉 **6. Step-by-Step Planned Research Methodology**

Subsection:

👉 **6.6 Step 6 — Construct a five-path pool for each source-destination pair and vehicle category**

Page:

👉 Around **Page 5–6**

📌 Exact idea written

From the document:
72482448-e8b4-442d-9599-bd97da2…

- A pool is defined as:

  > a group of five alternative paths for the same source-destination pair **and vehicle category**

- It also says:

  > This pool must remain category-aware

- And:

A truck pool, a bus pool, and a small-vehicle pool for the same source-destination pair may differ substantially

---

📖 Interpretation

This means:

- Pool is **NOT generic**
- It is:
  - **pair-specific (s → d)**
  - **category-specific (C1, C2, C3, C4)**

So:

- Same source-destination → **different pools for different vehicle types**

✔️ Final conclusion

✔️ **YES — your document correctly includes this concept**
✔️ It is written **explicitly and clearly** (good for supervisor)

If you want, I can check:
👉 "Is pool size (5 paths) clearly mentioned or not?"

Sources

> If a road is blocked, does it affect all the cars and source destinations that are using that section of edge?

Yes — **this concept is present in your document**, but again it is written in a **distributed explanatory way**, not as a single sentence.

✅ Answer: YES

📍 Where it appears

Section:

👉 6. Step-by-Step Planned Research Methodology

🔹 Step 9 — Identify which trips are affected when a disruption appears

Page: ~Page 5–6

This part clearly states:

- when a disruption occurs:
  - identify the **blocked edge**
  - determine **which source-destination pairs rely on that corridor**
  - identify **affected trips**

📌 Meaning (very important)

This directly implies:

✔️ If a road (edge) is blocked →
➡️ **all vehicles (cars) using that edge are affected**
➡️ **all source-destination pairs that depend on that edge are affected**

🔹 Also supported in:

Section 2 — Central Research Question

- mentions disruption affecting routing behavior

Step 8 — Monitor disruption

- defines types of disruptions (blocked edge, broken link, etc.)

📖 Interpretation

Your sentence:

> "If a road is blocked, does it affect all the cars and source destinations that are using that section of edge?"

✔️ In your document, this is represented as:

- identifying **all affected S-D pairs**
- and then applying rerouting (Case 1, 2, 3)

✔️ Final conclusion

✔️ **YES — the concept is clearly included**
✔️ It is handled properly in **Step 9 (very important section)**

If you want, I can check:
👉 whether the document explicitly mentions **"vehicles currently on the edge" vs "future trips"** (this is a deeper concept your supervisor may ask).

Sources

If it contains that if source-destination is blocked, then new route design by maintaining the same algo, and in designing the route, you should avoid that a route was blocked.

Yes — **this is present in the document**.

It appears mainly in the **step-by-step methodology** where the document explains what happens after a disruption and how rerouting is done while avoiding the blocked edge.

Where it appears:

- **Section 6.9 — Identify which trips are affected when a disruption appears**
  - says the system must first identify the **blocked edge or affected corridor** and the affected source-destination pairs.
- **Section 6.10 — Evaluate Case 1: source-based rerouting**
  - says that once a **blocked edge is identified**, the edge is **removed from the usable network** and a **new feasible route is planned again from the original source**.
- **Section 6.11 — Evaluate Case 2: current-position rerouting**
  - says the system should **increase local weights around the congested or blocked area** and **replan from the current node**, which also means the new route is designed while respecting the disruption.
- **Section 6.12 — Evaluate Case 3: pool-based rerouting**
  - says the system queries the precomputed pool, **removes infeasible candidates**, and selects the best remaining feasible path. That means blocked-route candidates are excluded.

So the answer is:

- **Yes**, the document contains that when a route/edge is blocked, a **new route is designed using the same framework**
- and the new design must **avoid the blocked route/edge**

One careful note:

- the document does **not say "same algorithm" in one exact sentence**
- but it does say the rerouting is done within the **same planned framework** using **Case 1, Case 2, and Case 3** after disruption.

Sources

Does it contain error case 1 and case 2? In case 1, does it contain also that if it removes the block edge from its network? And in case 2, does it contain that if it changes the weight for the block edge?

Yes — **both Case 1 and Case 2 are in the document**, and the specific actions you asked about are also included.

For **Case 1**:

- **Section: 6.10 Step 10 — Evaluate Case 1: source-based rerouting**
- **Page:** around **page 6**
- **Paragraph:** the full paragraph under Step 10
- **What it says:** once a **blocked edge is identified**, "the edge is removed from the usable network and a new feasible route is planned again from the original source."

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So:

- **Yes**, it contains **Case 1**
- **Yes**, it contains that the **blocked edge is removed from the network**

For **Case 2**:

- **Section: 6.11 Step 11 — Evaluate Case 2: current-position rerouting**
- **Page:** around **page 6**
- **Paragraph:** the full paragraph under Step 11
- **What it says:** the system should "**increase local weights around the congested or blocked area** and replan from the current node or local current position."

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So:

- **Yes**, it contains **Case 2**
- **Yes**, it contains that the **weight is changed/increased around the blocked area**

One careful note:

- it says **increase local weights around the blocked or congested area**
- it does **not say only the single blocked edge weight changes**
- so the wording is a bit broader than just "change the blocked edge weight."

72482448-e8b4-442d-9599-bd97da2…

So the final answer is:

- **Case 1:** yes, blocked edge is removed
- **Case 2:** yes, weights are increased around the blocked/congested area

Sources

Does it contain individual source destination without broken link, and also does it contain links maybe broken with the probability?(in time of pool making)

Yes — **both of these ideas are present in your document**, and I'll point them precisely.

✅ 1. Individual source–destination without broken link (steady-state)

✔️ YES — present

📍 Where:

- **Section: 6.5 Step 5 — Build category-wise feasible route options under steady-state conditions**
- **Page:** ~Page 5

📌 What it says:

- routes are built **before any sudden block is introduced**
- this is the **normal / steady-state condition**

  "before any sudden block is introduced, the system should determine which routes are feasible..."

72482448-e8b4-442d-9599-bd97da2…

👉 Meaning:

- routing is done for **individual source–destination pairs**
- under **no broken link / no disruption condition**

✅ 2. Links may be broken with probability (during pool making)

✔️ YES — present (very important concept)

📍 Where:

- **Section: 6.3 Step 3 — Classify all roads...**
- **Page:** ~Page 4–5

📌 What it says:

- mentions **fragile or historically sensitive corridors**
- can be assigned **higher disruption priors**

  "fragile or historically sensitive corridors may later be assigned higher disruption priors"

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📍 Also in:

- **Section: 6.6 Step 6 — Pool construction**

📌 What it says:

- during pool building:

  "fragile links may receive higher prior caution… so that the pool does not rely too heavily on corridors that are likely to become blocked"
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📖 Interpretation

This means:

- links can have **probability / risk of being broken**
- this is considered **during pool construction (steady-state)**
- pool avoids **high-risk links**

✔️ Final conclusion

Concept	Present or Not
Individual S-D routing without broken link	✅ YES
Broken-link probability / disruption prior in pool making	✅ YES

If you want, I can check next:
👉 "Is Shahbagh / Science Lab example correctly handled as probability (not fixed)?"

Sources

> Does it contain any information that this text contains any information that if this work is done by anyone else?(such as in reference)

Yes — **your document does contain information that this work is connected to previous works (other researchers), but in a proper academic way through references and future work—not as "this work is already done by others."**

✅ Answer: YES (but carefully framed)

📍 Where it appears

Section:

👉 **9. Detailed Future Work Informed by the Uploaded Literature**
👉 **Page:** ~Page 7–8

📌 What it says:

- your work is **informed by existing research papers**
- and you are planning to **extend or adapt ideas**

Example idea:

> "The uploaded literature suggests several next-step refinements..." 72482448-e8b4-442d-9599-bd97da2…

📍 References section

Section:

👉 11. References

- Lists multiple research papers (IEEE format)
- These are **previous works by other researchers**

📖 Interpretation (very important)

Your document:

- ❌ does **NOT say**: "this work is already done by others"
- ✔️ does say:
  - your work is **inspired by / informed by prior research**
  - your work is **an extension / new framework**

✔️ Academic meaning

This is exactly correct academically:

- You show:
  - awareness of existing work
  - connection to literature
- But you still present:
  - **your framework as new / proposed**

⚠️ Important note (for your supervisor)

Your document is **safe** because:

- it avoids claiming others already solved your exact problem

- it keeps your work as:
  👉 **“planned method” + “future work” + “hypothesis”**

✔️ Final conclusion

✔️ **YES — references to other work are included**
✔️ But they are used as:

- **supporting literature**
- **future work inspiration**
  ❌ Not as “this work already exists”

If you want, I can check:
👉 “Does the document clearly show your novelty (what is new in your work)?”

Sources

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Document

read it and tell me problem scoe briefly

Problem scope, briefly:

The work studies **disruption-resilient routing in Dhaka**: when a road is suddenly blocked, broken, or closed, vehicles should be rerouted **quickly** through **feasible alternative paths**. The scope is broader than normal shortest-path planning because it must consider **road hierarchy, road width, road condition, vehicle-category feasibility, pair-level routing, network-level interaction, sudden disruption handling, and comparison of three rerouting cases**. It also covers **all relevant source-destination pairs**, not just one example.

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In simple words:

- not every vehicle can use every road
- routes must be selected category-wise
- disruption affects multiple trips
- the study compares different rerouting strategies to see which responds better after a blockage.

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Sources